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**16.COMPARE DIFFERENT TYPES CLASSIFICATION ALGORITHMS  
AND EVALUATE THEIR PERFORMANCE.**

**1. Student Result Classification – Decision Tree**

| Study Hours | Attendanc<br>e | Result |
|-------------|----------------|--------|
| 2           | 60             | Fail   |
| 3           | 65             | Fail   |
| 4           | 70             | Fail   |
| 5           | 75             | Pass   |
| 6           | 80             | Pass   |
| 7           | 85             | Pass   |
| 8           | 90             | Pass   |
| 9           | 95             | Pass   |

**PROGRAM**

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
df = pd.read_csv("student.csv")
X = df[["Study Hours", "Attendance"]]
y = df["Result"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)
```

```
model = DecisionTreeClassifier(random_state=1)
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Predicted:", pred)
print("Actual:", y_test.values)
print("Accuracy:", accuracy_score(y_test, pred))
```

## OUTPUT

Predicted: ['Pass' 'Fail' 'Pass']

Actual: ['Pass' 'Fail' 'Pass']

Accuracy: 1.0

Accuracy = 100%

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## 2. Loan Approval – KNN

| Income | Credit Score | Loan |
|--------|--------------|------|
| 8      | 780          | Yes  |
| 7      | 760          | Yes  |
| 6      | 720          | Yes  |
| 4      | 620          | No   |
| 3      | 580          | No   |
| 9      | 800          | Yes  |
| 5      | 650          | No   |
| 8      | 770          | Yes  |
| 4      | 600          | No   |
| 7      | 740          | Yes  |

## **PROGRAM**

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
df = pd.read_csv("loan.csv")
X = df[["Income", "Credit Score"]]
y = df["Loan"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Predicted:", pred)
print("Actual:", y_test.values)
print("Accuracy:", accuracy_score(y_test, pred))
```

## **OUTPUT**

Predicted: ['No' 'Yes' 'Yes']

Actual: ['No' 'Yes' 'Yes']

Accuracy: 1.0

Accuracy = 100%

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### 3. Disease Classification – Naïve Bayes

| Temperature | Heart Rate | Disease  |
|-------------|------------|----------|
| 39.0        | 110        | Positive |
| 38.5        | 108        | Positive |
| 37.0        | 78         | Negative |
| 39.2        | 115        | Positive |
| 36.8        | 75         | Negative |
| 38.8        | 112        | Positive |
| 37.2        | 80         | Negative |
| 39.1        | 114        | Positive |
| 36.7        | 74         | Negative |
| 38.9        | 111        | Positive |

#### PROGRAM

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score
df = pd.read_csv("disease.csv")
X = df[["Temperature", "Heart Rate"]]
y = df["Disease"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)
model = GaussianNB()
model.fit(X_train, y_train)
```

```

pred = model.predict(X_test)
print("Predicted:", pred)
print("Actual:", y_test.values)
print("Accuracy:", accuracy_score(y_test, pred))

```

## OUTPUT

Predicted: ['Negative' 'Positive' 'Positive']

Actual: ['Negative' 'Positive' 'Positive']

Accuracy: 1.0

Accuracy = 100%

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## 4. Employee Promotion – Logistic Regression

| <div> <div></div> <div></div> <div></div> <div></div> </div> Experience | Performance | Promotion |
|-------------------------------------------------------------------------|-------------|-----------|
| 10                                                                      | 95          | Yes       |
| 8                                                                       | 90          | Yes       |
| 7                                                                       | 88          | Yes       |
| 3                                                                       | 65          | No        |
| 2                                                                       | 60          | No        |
| 9                                                                       | 93          | Yes       |
| 4                                                                       | 70          | No        |
| 8                                                                       | 89          | Yes       |
| 2                                                                       | 58          | No        |
| 6                                                                       | 85          | Yes       |

## PROGRAM

```
import pandas as pd
```

```

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
df = pd.read_csv("employee.csv")
X = df[["Experience", "Performance"]]
y = df["Promotion"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Predicted:", pred)
print("Actual:", y_test.values)
print("Accuracy:", accuracy_score(y_test, pred))

```

## OUTPUT

Predicted: ['No' 'Yes' 'Yes']

Actual: ['No' 'Yes' 'Yes']

Accuracy: 1.0

Accuracy = 100%

## 5. Fruit Classification – Random Forest

| Weight | Sweetness | Fruit  |
|--------|-----------|--------|
| 150    | 90        | Apple  |
| 160    | 88        | Apple  |
| 140    | 91        | Apple  |
| 120    | 70        | Orange |

|     |    |        |
|-----|----|--------|
| 125 | 72 | Orange |
| 130 | 74 | Orange |
| 155 | 89 | Apple  |
| 118 | 69 | Orange |
| 148 | 92 | Apple  |
| 122 | 71 | Orange |

## PROGRAM

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

df = pd.read_csv("fruit.csv")
X = df[["Weight", "Sweetness"]]
y = df["Fruit"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)

model = RandomForestClassifier(
    n_estimators=20, random_state=1)
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Predicted:", pred)
print("Actual:", y_test.values)
print("Accuracy:", accuracy_score(y_test, pred))
```

## OUTPUT

```
Predicted: ['Apple' 'Apple' 'Orange']
Actual:    ['Apple' 'Apple' 'Orange']
```

Accuracy: 1.0